

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 2.4: ACIDS, BASES AND SALTS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION ASSESSMENT — STUDENT TASK GUIDE

Driving Question: Why do some substances from our everyday Kenyan environment — lake water, lemon juice, wood ash, cooking salt, antacids — react so differently with metals, food and living things, and how can understanding their acid, base or salt chemistry help us use them safely and sustainably?

WHAT IS A FINAL EXPLANATION?

A Final Explanation is your chance to show everything you have learned across all 8 lessons by answering the driving question fully and scientifically. Think of it as writing a detailed scientific report for a fellow student who missed the entire unit — someone smart, curious, and Kenyan, who wants to understand the chemistry happening in their own kitchen, farm, and environment.

Your Final Explanation must do THREE things:

1. Use correct scientific vocabulary and concepts (acids, bases, salts, pH, neutralisation, hydrolysis, deliquescence, efflorescence, etc.)
2. Connect your explanation directly back to the anchoring phenomenon — Lake Magadi's soda crusts AND the damaged aluminium sufuria.
3. Show how the chemistry is useful and relevant to real Kenyan life — farming, cooking, health, industry, and environmental care.

FORMAT AND LENGTH

Your response must have FIVE clearly labelled sections. Each section has a specific prompt and a worked exemplar to guide you. Write in full sentences and paragraphs. Aim for 150–250 words per section. Use diagrams, equations, or tables where they strengthen your explanation — they do not replace written explanation.

BEFORE YOU BEGIN — SELF-CHECK CHECKLIST

Read through each prompt carefully. Before writing, tick off the following:

- ☐ I can define acid, base, and salt in my own words AND using particle/ion language.
- ☐ I can explain what pH means and give examples from the Kenyan phenomenon.
- ☐ I can write and explain a neutralisation equation.
- ☐ I can explain why the sufuria was damaged by lemon juice.
- ☐ I can explain what trona is, why it forms at Lake Magadi, and how it behaves in air.
- ☐ I can connect acid-base-salt chemistry to at least TWO real Kenyan uses or safety decisions.

SCORING

Your teacher will assess your Final Explanation using a 4-criteria rubric (Excellent / Proficient / Developing). Read the rubric BEFORE you write so you know exactly what is expected. Each criterion is equally weighted.

Section 1 — Classifying the Substances: Acids, Bases, and Salts

Explain what acids, bases, and salts are at the particle (ion) level. Then classify each of the following substances from our phenomenon and from Kenyan daily life, giving a reason for each classification: Lake Magadi water, lemon juice, wood ash solution, trona (sodium sesquicarbonate, $\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$), cooking salt (NaCl), and antacid tablets (e.g. magnesium hydroxide). Use pH values where you can, and explain what the pH scale tells us about strength and concentration of acids and bases.

Acids, bases, and salts are three categories of compounds that behave very differently because of the ions they produce in water. An acid releases hydrogen ions (H^+) in solution — the more H^+ ions present, the stronger and more corrosive the acid. A base releases hydroxide ions (OH^-) in solution. A salt is the ionic compound formed when an acid and a base react together, and it does not itself produce H^+ or OH^- directly (though some salts can hydrolyse in water to give a slightly acidic or basic solution).

The pH scale runs from 0 to 14 and measures the concentration of H^+ ions in a solution. A pH below 7 is acidic, pH 7 is neutral, and pH above 7 is alkaline (basic). A change of one pH unit means a tenfold change in H^+ concentration — so Lake Magadi water at around pH 10–11 is millions of times more alkaline than pure water.

Classifying our Kenyan substances:

- Lemon juice (pH $\approx$ 2–3): ACID. Lemons contain citric acid, which releases H^+ ions. This is why lemon juice tastes sour and reacted with the aluminium sufuria overnight.
- Wood ash solution (pH $\approx$ 9–11): BASE. Wood ash contains potassium carbonate (K_2CO_3), which reacts with water to produce OH^- ions. Kenyan farmers sometimes use wood ash to reduce soil acidity.
- Lake Magadi water (pH $\approx$ 10–11): STRONGLY ALKALINE BASE. It is rich in dissolved sodium carbonate (Na_2CO_3) and sodium bicarbonate (NaHCO_3), making it caustic.
- Trona ($\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$): SALT — specifically a basic salt, because it is formed from a strong base and a weak acid. When dissolved, it hydrolyses to give an alkaline solution.
- Cooking salt, NaCl: NEUTRAL SALT. Formed from the strong acid HCl and the strong base NaOH, it does not hydrolyse and gives pH 7 in solution.
- Antacid tablets ($\text{Mg}(\text{OH})_2$): BASE. They contain a base that neutralises excess stomach acid (HCl), relieving heartburn. This is a direct and safe use of acid-base chemistry in Kenyan homes.

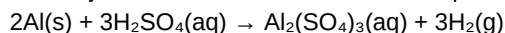
Section 2 — Neutralisation and Salt Formation: Explaining the Reactions

Explain the process of neutralisation using particle (ionic) language. Write and interpret at least TWO balanced chemical equations for neutralisation reactions — one must be directly relevant to the sufuria phenomenon (lemon juice + aluminium) and one must relate to a practical Kenyan context such as treating acidic soil with lime, or using antacids. Then explain what a salt is as a product of neutralisation, and discuss how different combinations of strong/weak acids and bases produce salts with different pH values in solution (salt hydrolysis).

Neutralisation is the reaction between an acid and a base in which H^+ ions from the acid combine with OH^- ions from the base to form water. At the particle level, the essential reaction is always: $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$. The remaining ions pair up to form a salt. The reaction releases heat (it is exothermic), which is why mixing strong acids and bases causes a temperature rise.

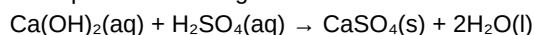
Example 1 — The damaged sufuria (acid attacking metal):

Lemon juice contains citric acid, but a simpler model uses the general reaction of an acid with aluminium metal:



In our case, the aluminium oxide protective layer on the sufuria reacts with citric acid. The H^+ ions from the lemon juice attack the Al_2O_3 coating and eventually pit the metal surface. This is why leaving lemon-water in the sufuria overnight was so damaging — prolonged contact allowed the acid to breach the protective layer.

Example 2 — Treating acidic farm soil in Kericho with agricultural lime (calcium hydroxide):



The base (lime) neutralises the acid in the soil. The salt produced, calcium sulphate, is relatively insoluble and harmless to crops. Tea farmers in Kericho apply lime to acidic volcanic soils to raise pH toward the neutral range that tea bushes prefer.

Salt hydrolysis: Not all salts give a neutral solution. Trona, formed from a strong base (NaOH) and a weak acid (H_2CO_3), hydrolyses in water to give an alkaline solution — this explains why Lake Magadi water is so basic. Ammonium chloride (NH_4Cl), formed from a weak base and a strong acid, hydrolyses to give an acidic solution. Cooking salt (NaCl) — strong acid + strong base — gives a neutral solution. Understanding salt hydrolysis explains why you must not assume all salts are safe just because they are 'neutral-sounding'.

Section 3 — Behaviour of Salts in Air: Deliquescence, Efflorescence, and Hygroscopy

The KICD key inquiry questions ask: 'Why are salts important in day-to-day life?' and 'How do salts behave when exposed to air?' Answer both questions together in this section. Explain the difference between deliquescence, efflorescence, and hygroscopy at the particle level. Use trona from Lake Magadi and the mineral scale inside the sufuria as your central examples. Then describe at least TWO practical consequences of these behaviours — one positive (useful) and one negative

Salts are enormously important in daily Kenyan life — from the NaCl we season ugali and nyama choma with, to the soda ash from Lake Magadi used in glass-making and detergents, to the fertiliser salts that help maize and beans grow in Rift Valley farms. But salts behave in fascinating and sometimes surprising ways when left exposed to the air, and understanding this helps us store and use them safely.

Hygroscopic salts absorb water vapour from the surrounding air without dissolving completely. Deliquescent salts go further — they absorb so much water vapour that they dissolve entirely in the absorbed water, forming a solution. Calcium chloride (CaCl_2) is a classic deliquescent salt: leave it in open air and it becomes a puddle. This happens because the ions in the salt attract polar water molecules from the air so strongly that they pull them out of the gas phase. Efflorescent salts do the opposite — they lose water of crystallisation to the air when the surrounding air is drier than the salt's equilibrium vapour pressure. Washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) is efflorescent: left in dry air, it loses its water molecules and crumbles into a white powder (Na_2CO_3 , anhydrous soda ash).

Trona ($\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$) from Lake Magadi is a hydrated salt. When harvested and left in dry highland air, it can effloresce, losing some of its water of crystallisation. This changes its texture and mass — important for industrial processing.

(problematic) — in a Kenyan context.	The white mineral scale inside the sufuria is left by hard water — dissolved calcium and magnesium salts (mainly CaCO_3 and MgCO_3) that precipitate when water is boiled. This is a negative consequence: scale is a poor heat conductor, wastes fuel, and can harbour bacteria. Positively, the hygroscopic nature of some salts is exploited in Kenya — silica gel packets (a hygroscopic material) protect electronics and medicines from humidity damage in Mombasa's coastal air. Knowing whether a salt is deliquescent or efflorescent tells you how to store it: keep deliquescent salts in airtight containers; store efflorescent salts away from very dry heat.
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Section 4 — Explaining the Anchoring Phenomenon: Lake Magadi and the Sufuria

Return directly to both images from the anchoring phenomenon. Using ALL the chemistry you have learned in this unit, write a full scientific explanation of: (a) why Lake Magadi's water is so alkaline, how trona forms, and why the lake can be both caustic and ecologically and commercially valuable; and (b) why the lemon juice damaged the aluminium sufuria overnight, what the white mineral scale is and how it got there, and what a household in Nairobi could do — using acid-base chemistry — to safely remove the scale. Make sure your explanation links acids, bases, salts, pH, neutralisation, and the behaviour of salts in air or water.	(a) Lake Magadi — The Chemistry of a Caustic but Valuable Lake: Lake Magadi sits in a closed basin in Kenya's Southern Rift Valley. Rain and groundwater leach through volcanic rocks rich in sodium minerals and carry dissolved sodium carbonate (Na_2CO_3) and sodium bicarbonate (NaHCO_3) into the lake. Because the lake has no river outlet, water only leaves by evaporation — so over thousands of years, the sodium salts have become enormously concentrated. Na_2CO_3 is the salt of a strong base (NaOH) and a weak acid ($\text{CO}_2/\text{H}_2\text{CO}_3$), so it hydrolyses in water to give OH^- ions, making the lake strongly alkaline (pH 10–11). Trona ($\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$) crystallises out at the surface as water evaporates, forming the thick white crusts we see in the photograph. Yet this 'caustic' lake supports flamingos: the pink-red colour comes from alkali-tolerant cyanobacteria and algae (<i>Spirulina</i>) that flamingos feed on — organisms specially adapted to high pH. Commercially, Tata Chemicals Magadi mines the trona and converts it to pure soda ash (Na_2CO_3), used in making glass, detergents, and paper across East Africa and beyond. The Maasai communities have long harvested the salts as a livestock mineral supplement. So the same extreme alkalinity that would burn unprotected skin is, in the right context, an ecological habitat and an economic resource. (b) The Damaged Sufuria — Acid Attack, Mineral Scale, and Safe Cleaning: Aluminium cooking pots are normally protected by a thin layer of aluminium oxide (Al_2O_3) on their surface. When lemon juice (citric acid, pH $\approx$ 2–3) was left in the sufuria overnight, the sustained contact of H^+ ions dissolved this protective oxide layer: $\text{Al}_2\text{O}_3(\text{s}) + 3\text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{Al}_2(\text{SO}_4)_3(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \text{ (general model)}$ Once the oxide layer is gone, the bare aluminium reacts further with the acid, causing the grey pitting seen in the photograph. The lesson: never leave acidic liquids (lemon juice, tomato sauce, tamarind) in an aluminium sufuria overnight. The white scale inside the pot is calcium carbonate (CaCO_3) and magnesium carbonate (MgCO_3) — salts dissolved in Nairobi's hard tap water that precipitate as solid scale when the water is boiled: $\text{Ca}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{CaCO}_3(\text{s})$. To remove this scale safely, a household can use a mild acid — white vinegar (ethanoic acid) or dilute lemon juice — applied for a SHORT time (not overnight!) and then rinsed thoroughly. The acid neutralises and dissolves the basic carbonate scale: $\text{CaCO}_3(\text{s}) + 2\text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{Ca}(\text{CH}_3\text{COO})_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$. The fizzing you see is CO_2 gas being released — a sign the reaction is working. This is acid-base neutralisation being used safely and practically in a Kenyan kitchen.
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Section 5 — Safe and Sustainable Use: Personal, Community, and Environmental Decisions

Reflect on what you have learned and explain how understanding acid, base, and	Understanding the chemistry of acids, bases, and salts is not just a classroom exercise — it is knowledge that protects people, improves livelihoods, and helps Kenya use its natural resources responsibly.
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salt chemistry empowers you, your family, your community, and Kenyan society to make safer and more sustainable decisions. Address at least THREE of the following contexts: (i) safe food preparation and cookware care in Kenyan households; (ii) sustainable soil management for Kenyan farmers; (iii) industrial use of Lake Magadi soda ash and environmental responsibility; (iv) safe use of cleaning products, antacids, or agricultural chemicals; (v) water quality and health in Kenyan communities. For each context, name a specific action that is informed by acid-base-salt chemistry.	(i) Safe food preparation and cookware care: Knowing that acids corrode aluminium means Kenyan households can protect their sufurias by rinsing them immediately after cooking acidic foods like tomato stew or lemon-marinated fish, and by never storing acidic liquids in them overnight. Using a stainless steel or enamel pot for highly acidic dishes is a chemistry-informed choice that saves money and prevents ingesting aluminium ions. (ii) Sustainable soil management: Kenyan smallholder farmers in areas like Kisii, Kericho, and the Central Highlands often face acidic soils (pH below 5.5) caused by heavy rainfall leaching basic minerals. Applying agricultural lime (CaO or Ca(OH)_2) neutralises soil acidity, raises pH, makes nutrients like phosphorus available to plants, and improves maize and bean yields — a direct application of neutralisation chemistry. Over-liming, however, can raise pH too high, harming crops — so understanding pH ranges helps farmers apply the right amount. (iii) Industrial and environmental responsibility at Lake Magadi: Tata Chemicals Magadi's soda ash operation is economically vital, but mining must be balanced with protecting the lake ecosystem and the communities who depend on it. Understanding that the lake's alkalinity is maintained by ongoing volcanic groundwater input — not just by the existing trona deposits — means that responsible harvesting rates can be calculated. Acid-base chemistry also helps engineers treat any acidic industrial effluent before it is released, preventing further environmental damage. In summary, acid-base-salt chemistry is woven into the fabric of Kenyan daily life — from the ugali pot on the jiko to the flamingos at Lake Magadi. The more deeply you understand it, the better equipped you are to make decisions that keep your family healthy, your farm productive, and your environment protected for the next generation.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific Accuracy and Use of Vocabulary	All scientific concepts (acids, bases, salts, pH, neutralisation, hydrolysis, deliquescence, efflorescence) are defined correctly at the ion/particle level. Balanced chemical equations are written and interpreted accurately. Subject-specific vocabulary is used consistently and correctly throughout all five sections.	Most scientific concepts are defined correctly. At least one balanced equation is written with minor errors that do not affect overall meaning. Key vocabulary is mostly used correctly, with one or two imprecise terms.	Some scientific concepts are present but contain significant errors or misconceptions (e.g. confusing acids and bases, incorrect pH scale direction). Equations are missing or substantially incorrect. Vocabulary is used sparingly or incorrectly.
Explanation of the Anchoring Phenomenon	Both the Lake Magadi phenomenon and the damaged sufuria are fully and accurately explained using multiple chemistry concepts (pH, salt hydrolysis, acid-metal reactions, mineral scale formation). The explanation is coherent, logically sequenced, and explicitly links causes to observations in the photographs. A practical, chemistry-based solution (e.g. scale removal) is correctly described.	Both phenomena are addressed and mostly explained correctly. The chemistry used is relevant but may rely on only one or two concepts rather than integrating multiple ideas. The practical solution is mentioned but may lack chemical detail.	Only one of the two phenomena is addressed, or both are mentioned but without meaningful chemical explanation. The response describes observations (e.g. 'the pot turned grey') without explaining the underlying chemistry.

Connection to Kenyan Contexts and Real-Life Application	At least three distinct Kenyan real-life contexts are addressed (e.g. soil liming in Kericho, soda ash industry at Magadi, sufuria care in Nairobi, antacid use, water hardness). Each context includes a specific, chemistry-informed action or decision. Kenyan scientists, places, foods, or industries are referenced naturally and accurately.	Two Kenyan contexts are addressed with chemistry-informed actions. References to Kenyan places or examples are present but may be generic rather than specific. The link between chemistry and the real-life decision is mostly clear.	One or fewer Kenyan contexts are addressed. Real-life connections are vague or limited to restating information from class without showing how chemistry informs decisions. Few or no Kenyan-specific references are made.
Integration of Acid-Base-Salt Concepts Across the Unit (Synthesis)	The Final Explanation shows clear synthesis across all major sub-topics of the unit: classification of acids/bases/salts, pH, neutralisation, salt formation and hydrolysis, and behaviour of salts in air (deliquescence/efflorescence/hygroscopy). Ideas from different lessons are woven together cohesively rather than listed in isolation. Section 4 in particular draws on content from Sections 1–3.	Most major sub-topics are present and mostly connected. Some sections feel isolated (e.g. the section on salt behaviour in air is not linked back to the phenomenon), but the overall explanation shows reasonable integration of unit content.	Sub-topics are treated in isolation with little or no connection between sections. Major sub-topics (e.g. deliquescence/efflorescence, or salt hydrolysis) are missing or only briefly mentioned. The response reads as a list of definitions rather than a coherent explanation.
Clarity, Structure, and Scientific Communication	All five sections are clearly labelled and fully addressed. Writing is clear, logically sequenced, and accessible to a fellow student. Scientific evidence and reasoning are explicitly stated (e.g. 'because...', 'this means that...', 'the evidence for this is...'). Diagrams, equations, or tables are used appropriately to support — not replace — written explanation.	All five sections are present and mostly addressed. Writing is generally clear but occasionally unclear or jumps between ideas without explanation. Reasoning is present but sometimes implicit rather than explicit. Diagrams or equations are included but may not be fully explained in the text.	One or more sections are missing or very brief. Writing is difficult to follow in places, with unclear links between ideas. Little or no use of scientific reasoning language ('because', 'therefore', 'this shows'). Diagrams or equations, if present, are unexplained.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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